# Geometry, Triangles & Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • [FRQ] • Target: Parallel Lines & Transversal Angle Relationships

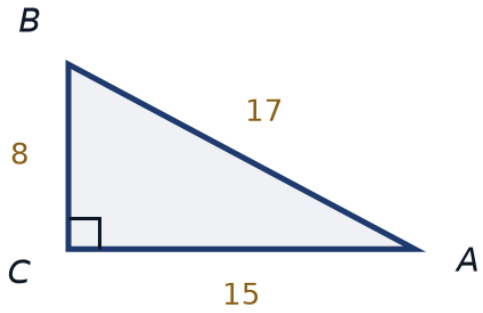
Two parallel lines $\ell_1 \parallel \ell_2$ are intersected by a transversal $t$. If a pair of consecutive interior (co-interior) angles have measures $(3x + 15)^\circ$ and $(2x - 5)^\circ$, find $x$ and the measure of both angles.

**2** [Foundational] • [FRQ] • Target: Triangle Angle-Sum & Exterior Angle Theorem

In $\triangle ABC$, the exterior angle at vertex $C$ measures $134^\circ$, and the interior angle $\angle A$ measures $58^\circ$. Find $\angle B$ and $\angle ACB$.

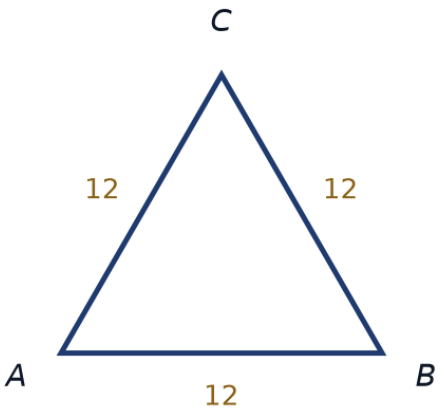
**3** [Foundational] • [FRQ] • Target: Pythagorean Theorem & Right-Triangle Trig Ratios

In right triangle $ABC$ shown below, $\angle C = 90^\circ$, $AC = 15$, and $BC = 8$. Find (a) the hypotenuse $AB$, (b) $\sin A$, (c) $\cos A$, and (d) $\tan B$.




**4** [Foundational] • [FRQ] • Target: Special Right Triangles ($30^{\circ}\mathrm{-}60^{\circ}\mathrm{-}90^{\circ}$)

Find the exact perimeter and area of an equilateral triangle whose altitude has length $6\sqrt{3}$ cm.




**5** [Foundational] • [FRQ] • Target: Special Right Triangles ($45^{\circ}\mathrm{-}45^{\circ}\mathrm{-}90^{\circ}$)

A square has diagonal of length $10\sqrt{2}$ m. Find its side length, perimeter, and area.

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**6** [Foundational] • [FRQ] • Target: Triangle Inequality Theorem

Two sides of a triangle have lengths $7$ and $12$. State the strict inequality for all possible lengths $x$ of the third side, and find how many integer values $x$ can take.

**7** [Foundational] • [FRQ] • Target: Isosceles Triangle Properties

In $\triangle PQR$, $PQ = PR$ (isosceles) and the vertex angle $\angle P$ measures $44^\circ$. Find the measure of each base angle $\angle Q$ and $\angle R$.

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## Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

**8** [Intermediate] • [FRQ] • Target: Similar Triangles & Parallel Cross-Sections

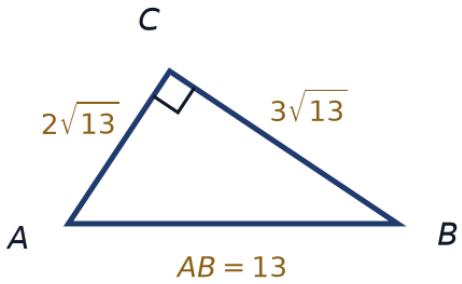
In $\triangle ABC$, segment $DE$ is parallel to base $BC$ with $D$ on $AB$ and $E$ on $AC$. Given $AD = 6$, $DB = 9$, and $DE = 8$, find the length of $BC$ and the ratio of the area of $\triangle ADE$ to the area of $\triangle ABC$.

**9** [Intermediate] • [FRQ] • Target: Complementary Angle Identity $\sin\theta = \cos(90^{\circ}-\theta)$

In right triangle $DEF$ with $\angle F = 90^\circ$, $\sin D = \frac{5}{13}$. Find $\cos E$, $\cos D$, and $\tan E$.

**10** [Intermediate] • [FRQ] • Target: Geometric Mean Theorems in Right Triangles

In right triangle $ABC$, the altitude $CD$ is drawn to the hypotenuse $AB$. If $AD = 4$ and $DB = 9$, find the length of the altitude $CD$ and both legs $AC$ and $BC$.



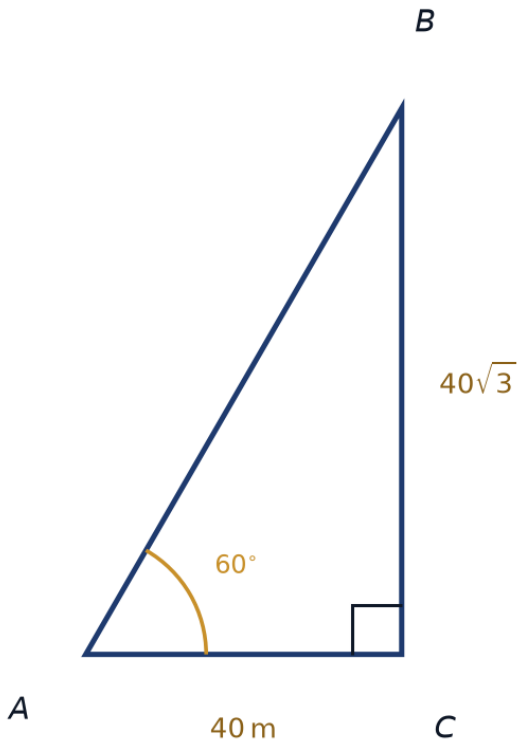

**11** [Intermediate] • [FRQ] • Target: Interior & Exterior Angles of Regular Polygons

Find the measure of one interior angle and one exterior angle of a regular $15$-sided polygon (pentadecagon).

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**12** [Intermediate] • [FRQ] • Target: Angles of Elevation & Depression

A surveyor stands $40\,\text{m}$ from the base of a vertical radio tower on level ground. The angle of elevation to the top of the tower is $60^\circ$, and the surveyor's eye level is $1.6\,\text{m}$ above the ground. Find the exact height of the tower and its decimal approximation to two decimal places.

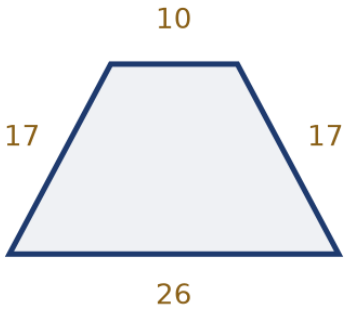



**13** [Intermediate] • [FRQ] • Target: Triangle Angle Bisector Theorem

In $\triangle ABC$, the angle bisector of $\angle A$ meets side $BC$ at point $D$. If $AB = 10$, $AC = 15$, and $BC = 20$, find the exact lengths of $BD$ and $DC$.

**14** [Intermediate] • [FRQ] • Target: Isosceles Trapezoids & Pythagorean Decomposition

An isosceles trapezoid has parallel bases of lengths $10\,\text{cm}$ and $26\,\text{cm}$ and non-parallel legs of length $17\,\text{cm}$. Find the height $h$, the area, and the length of each diagonal of the trapezoid.




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## Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging] • [FRQ] • Target: Coordinate Triangle Geometry & Centroid • Cross-Lesson: Topic 1.01 (Coordinate Geometry)**

In $\triangle ABC$, the vertices are $A(1, 2)$, $B(9, 2)$, and $C(5, 8)$. Verify that $\triangle ABC$ is isosceles, find the length of the altitude from $C$ to $AB$, and determine the coordinates of the centroid $G$.

**16** **[Challenging] • [FRQ] • Target: Algebraic Right Triangles via Quadratics • Cross-Lesson: Topic 3.01 (Quadratic Equations)**

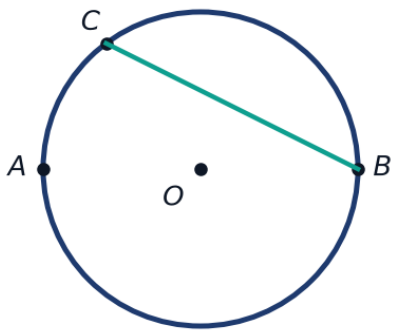
In a right triangle, one leg is $7\,\text{cm}$ longer than the other leg, and the hypotenuse is $17\,\text{cm}$. Set up and solve a quadratic equation to find the lengths of the two legs and the sine of the smaller acute angle.

**17** **[Challenging] • [FRQ] • Target: Law of Cosines & Trigonometric Triangle Area • Cross-Lesson: Topic 9.01 (Trigonometry)**

In $\triangle ABC$, $AB = 7$, $BC = 8$, and $\angle B = 60^\circ$. Use the Law of Cosines to find the exact length of side $AC$ and the exact area of $\triangle ABC$.

**18** **[Challenging] • [FRQ] • Target: Thales' Theorem & Inscribed Right Triangles • Cross-Lesson: Topic 9.01 (Circles & Inscribed Angles)**

Triangle $ABC$ is inscribed in a circle of diameter $AB = 20\,\text{cm}$. If $\cos A = \frac{3}{5}$, find the lengths of chords $AC$ and $BC$ and the area of the region inside the circle but outside $\triangle ABC$.




**19** **[Challenging] • [FRQ] • Target: Inscribed Square via Similar Right Triangles • Cross-Lesson: Topic 5.01 (Proportions) & Topic 8.01 (Area)**

A square is inscribed inside a right triangle with legs $6$ and $12$ so that one vertex of the square coincides with the right-angle vertex of the triangle and the opposite vertex lies on the hypotenuse. Find the side length $s$ of the square.

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**20** [Challenging] • [FRQ] • Target: Orthogonal Medians & Centroid Area Partition • Cross-Lesson: Topic 8.01 (Area Decomposition)

In $\triangle ABC$, medians $AD$ and $BE$ are perpendicular to each other and have lengths $AD = 9$ and $BE = 12$. Using the centroid ratio $2:1$, find the length of side $AB$ and the area of $\triangle ABC$.

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## Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

**21** [Foundational] • [MCQ] • Target: Triangle Angle-Sum Theorem

In $\triangle ABC$, $\angle A = 42^\circ$ and $\angle B = 73^\circ$. What is the measure of $\angle C$?

**a.** $55^\circ$

**b.** $65^\circ$

**c.** $75^\circ$

**d.** $115^\circ$

**22** [Foundational] • [MCQ] • Target: Pythagorean Triples

In a right triangle, the legs have lengths $9$ and $12$. What is the length of the hypotenuse?

**a.** 13

**b.** 15

**c.** 18

**d.** 21

**23** [Foundational] • [MCQ] • Target: Complementary Trig Identities

If $\sin(3x - 10)^\circ = \cos(2x + 25)^\circ$, where both angles are acute, what is the value of $x$?

**a.** 13

**b.** 15

**c.** 17

**d.** 21

**24** [Foundational] • [MCQ] • Target: Polygon Interior Angle Sum

What is the sum of the interior angles of a convex octagon ($8$ sides)?

**a.** $900^\circ$

**b.** $1080^\circ$

**c.** $1260^\circ$

**d.** $1440^\circ$

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**25** [Foundational] • [MCQ] • Target: Proportions in Similar Triangles

Triangles $ABC$ and $DEF$ are similar, with $AB$ corresponding to $DE$. If $AB = 6$, $BC = 10$, and $DE = 9$, what is the length of $EF$?

**a.** 12

**b.** 13

**c.** 15

**d.** 18

**26** [Foundational] • [MCQ] • Target: Special Right Triangle Ratios

In a $30^\circ\text{–}60^\circ\text{–}90^\circ$ right triangle, the side opposite the $30^\circ$ angle has length $7$. What is the length of the side opposite the $60^\circ$ angle?

**a.** $7\sqrt{2}$

**b.** $7\sqrt{3}$

**c.** 14

**d.** $\frac{7\sqrt{3}}{2}$

**27** [Foundational] • [MCQ] • Target: Alternate Interior & Supplementary Angles

In the figure below, lines $\ell$ and $m$ are parallel. If $\angle 1 = 128^\circ$, what is the measure of its alternate interior angle's supplement?

**a.** $38^\circ$

**b.** $52^\circ$

**c.** $62^\circ$

**d.** $128^\circ$

**28** [Foundational] • [MCQ] • Target: Testing the Triangle Inequality

Which of the following sets of side lengths CANNOT form a triangle?

**a.** $4, 5, 8$

**b.** $6, 6, 11$

**c.** $5, 9, 14$

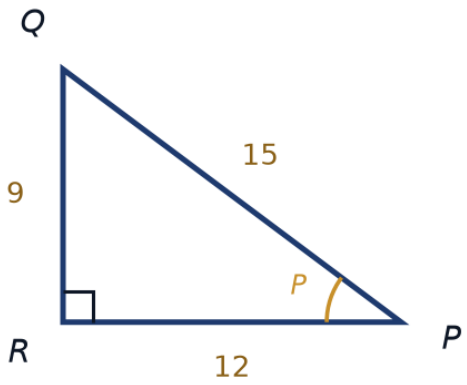
**d.** $7, 10, 12$

## Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

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**29** **[Intermediate]** • **[MCQ]** • **Target: Perimeter from Trigonometric Ratio**

In right triangle $PQR$ shown below, $\angle R = 90^\circ$, $PR = 12$, and $\tan P = \frac{3}{4}$. What is the perimeter of $\triangle PQR$?




**a.** 28

**b.** 32

**c.** 36

**d.** 48

**30** **[Intermediate]** • **[MCQ]** • **Target: Area-to-Perimeter Ratio in Similar Triangles**

Two similar triangles have perimeters in the ratio $3 : 7$. If the area of the smaller triangle is $27\,\text{cm}^2$, what is the area of the larger triangle?

**a.** $63\,\text{cm}^2$

**b.** $108\,\text{cm}^2$

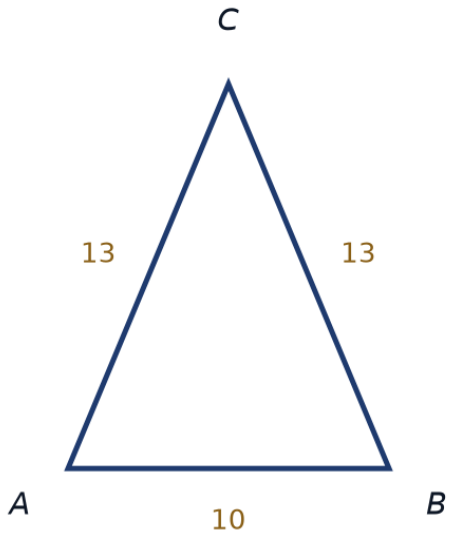
**c.** $147\,\text{cm}^2$

**d.** $189\,\text{cm}^2$

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**31** [Intermediate] • [MCQ] • Target: Altitude & Area of Isosceles Triangles

In an isosceles triangle with two sides of length $13$ and base of length $10$, what is the area of the triangle?




**a.** 60

**b.** 65

**c.** 120

**d.** 130

**32** [Intermediate] • [MCQ] • Target: Finding Number of Sides from Interior Angle

If each interior angle of a regular polygon measures $165^\circ$, how many sides does the polygon have?

**a.** 18

**b.** 20

**c.** 24

**d.** 30

**33** [Intermediate] • [MCQ] • Target: Geometric Mean Theorem for Right-Triangle Legs

In right triangle $ABC$ with $\angle C = 90^\circ$, altitude $CD$ is drawn to hypotenuse $AB$. If $AC = 10$ and $AD = 5$, what is the length of $AB$?

**a.** 15

**b.** 20

**c.** 25

**d.** $5\sqrt{5}$

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**34** [Intermediate] • [MCQ] • Target: Sliding Ladder Pythagorean Word Problems

A $25$-foot ladder leans against a vertical wall with its base $7$ feet from the wall. If the top of the ladder slides down the wall by $4$ feet, how many feet does the base of the ladder slide away from the wall?

**a.** $4$

**b.** $7$

**c.** $8$

**d.** $15$

**35** [Intermediate] • [MCQ] • Target: Midsegment Theorem & Area Partition

In $\triangle ABC$, $D$ is the midpoint of $AB$ and $E$ is the midpoint of $AC$. If the area of trapezoid $DBCE$ is $45$, what is the area of $\triangle ABC$?

**a.** $54$

**b.** $60$

**c.** $75$

**d.** $90$

**36** [Intermediate] • [MCQ] • Target: Pythagorean Trigonometric Identities

If $\theta$ is an acute angle such that $\sin\theta = 3\cos\theta$, what is the value of $\sin\theta\cos\theta$?

**a.** $\frac{1}{10}$

**b.** $\frac{3}{10}$

**c.** $\frac{1}{3}$

**d.** $\frac{3}{5}$

**37** [Intermediate] • [MCQ] • Target: Altitude Decomposition of Oblique Triangles

In $\triangle ABC$, $\angle A = 30^\circ$, $\angle C = 45^\circ$, and $AB = 12$. What is the exact length of $BC$?

**a.** $6\sqrt{2}$

**b.** $6\sqrt{3}$

**c.** $6\sqrt{6}$

**d.** $12\sqrt{2}$

## Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • [MCQ] • Target: Inradius of a Right Triangle • Cross-Lesson: Topic 9.01 (Circles & Tangent Segments)

What is the radius of the circle inscribed in a right triangle with legs $9$ and $40$?

**a.** $3.5$

**b.** $4$

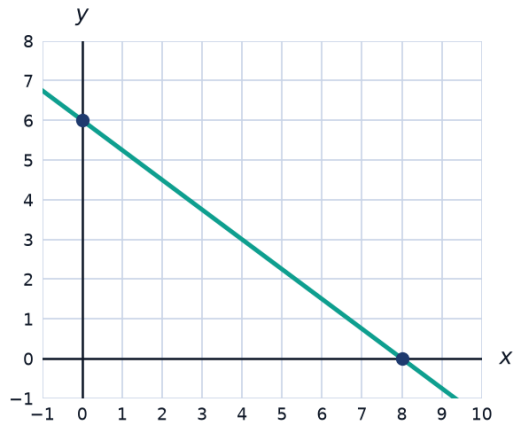
**c.** $4.5$

**d.** $5$

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**39** [Challenging] • [MCQ] • Target: Slope as Tangent & Trigonometric Ratios of Lines • Cross-Lesson: Topic 1.01 (Linear Intercepts & Slope)

In the $xy$-plane, line $y = -\frac{3}{4}x + 6$ forms a right triangle with the positive coordinate axes. What is the value of $\sin\theta$, where $\theta$ is the acute angle the line makes with the $x$-axis?




**a.** $\frac{3}{5}$

**b.** $\frac{4}{5}$

**c.** $\frac{3}{4}$

**d.** $\frac{4}{3}$

**40** [Challenging] • [MCQ] • Target: Heron's Formula & 13-14-15 Triangle Altitude • Cross-Lesson: Topic 8.01 (Area of Triangles)

In $\triangle ABC$, sides $a, b, c$ have lengths $13, 14, 15$. What is the length of the altitude to the side of length $14$?

**a.** 10

**b.** 11

**c.** 12

**d.** 12.5

**41** [Challenging] • [MCQ] • Target: Algebraic Squaring with $\sin^2\theta + \cos^2\theta = 1$ • Cross-Lesson: Topic 3.01 (Quadratic Identities)

If $\sin\theta + \cos\theta = \frac{7}{5}$ for an acute angle $\theta$, what is the value of $\sin\theta\cos\theta$?

**a.** $\frac{12}{25}$

**b.** $\frac{24}{25}$

**c.** $\frac{7}{25}$

**d.** $\frac{12}{5}$

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**42** [Challenging] • [MCQ] • Target: Isoperimetric Area Ratio of Regular Polygons • Cross-Lesson: Topic 5.01 (Ratios) & Topic 8.01 (Polygon Area)

A regular hexagon and an equilateral triangle have the exact same perimeter. What is the ratio of the area of the regular hexagon to the area of the equilateral triangle?

**a.** $1:1$

**b.** $3:2$

**c.** $2:1$

**d.** $4:3$

**43** [Challenging] • [MCQ] • Target: Incenter Angle Formula • Cross-Lesson: Topic 2.01 (Linear Angle Systems)

In $\triangle ABC$, angle bisectors of $\angle B$ and $\angle C$ intersect at the incenter $I$. If $\angle A = 68^\circ$, what is the measure of $\angle BIC$?

**a.** $112^\circ$

**b.** $124^\circ$

**c.** $134^\circ$

**d.** $146^\circ$

**44** [Challenging] • [MCQ] • Target: Crossed-Ladders Similar Triangles Theorem • Cross-Lesson: Topic 10.01 (Harmonic & Rational Equations)

Two poles of heights $10\,\text{m}$ and $15\,\text{m}$ stand vertically on level ground $20\,\text{m}$ apart. A taut wire connects the top of each pole to the base of the opposite pole. At what height above the ground do the two wires intersect?

**a.** $5\,\text{m}$

**b.** $6\,\text{m}$

**c.** $6.25\,\text{m}$

**d.** $7.5\,\text{m}$

**45** [Challenging] • [MCQ] • Target: Coordinate Triangle Area via Bounding Box / Shoelace • Cross-Lesson: Topic 1.01 (Coordinate Plane)

What is the area of the triangle with vertices $(0, 0)$, $(6, 2)$, and $(3, 8)$ in the $xy$-plane?

**a.** $18$

**b.** $21$

**c.** $24$

**d.** $27$

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